

Project Initialization and Planning Phase

Date	04 July 2026
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Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

If a farmer need any suggestion or advice there is no one to suggest them.as mostly they are not educated It is harder to scientifically understand the conditions of soil ,seasons and regoin based cultivation.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A smallholder farmer in Andhra Pradesh growing 3–5 acres of crops.	Decide which crop to plant next season for maximum yield without expensive soil tests.	Crop selection is based on guesswork or outdated advice from neighbours.	Soil testing labs charge ₹500–₹2000 per test and are far from rural villages.	Anxious, financially stressed, and uncertain about the coming season.
PS-2	An agricultural extension officer or agronomist.	Quickly assess whether a farmer's current soil and climate conditions suit a chosen crop.	Assessment is manual, time-consuming, and relies on fragmented data sources.	No unified digital tool evaluates N, P, K, temperature, humidity, pH, and rainfall together.	Overwhelmed and unable to serve all farmers efficiently.
PS-3	An agricultural researcher or government policy maker.	Analyse crop-environment relationships across regions and seasons to design evidence-based strategies.	Data is siloed, in non-standard formats, and not easily filterable or exportable.	No interactive analytics platform exists that combines crop, climate, and nutrient data.	Frustrated and unable to produce timely policy recommendations.